Multi-Agent Task Allocation with Deep Learning

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Deep Learning – Winter Term 2026/2027
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***Abstract*—Multi-Agent Task Allocation (MATA) is an important problem in multi-agent systems, where several autonomous agents must coordinate their actions to complete tasks efficiently. Task allocation becomes more difficult in dynamic environments, where new tasks may appear, agents have limited information, and some tasks require cooperation. Traditional approaches use optimization algorithms, auction-based methods, and predefined rules to assign tasks. Recent advances in Deep Reinforcement Learning (DRL) allow agents to learn task allocation strategies through interaction with their environment.**

This paper examines the application of Deep Learning to dynamic multi-agent task allocation, focusing on decentralized coordination and coalition formation. The theoretical foundations of Multi-Agent Reinforcement Learning (MARL) are discussed, followed by a case study of MARL-MRT^{2}A, which combines Multi-Agent Proximal Policy Optimization (MAPPO) with a U-Net architecture. The approach is examined using the GyMRT^{2}A simulation environment. An experimental reproduction investigates how task allocation performance changes during training by evaluating policies at different training stages. The evaluation considers episodic reward and task completion, with the market-based Project Manager-oriented Coalition Formation Algorithm (PCFA) serving as a comparison baseline.

I. Introduction

Multi-agent systems consist of several autonomous agents that work in a shared environment to achieve individual or common goals. Each agent collects information about its surroundings through sensors and observations. Agents can exchange this information to coordinate their actions, avoid unnecessary work, and use available resources more efficiently. However, effective cooperation also requires deciding which agent should perform each task. This problem is known as Multi-Agent Task Allocation (MATA).

Task planning in multi-agent systems includes three main problems: task decomposition, task allocation, and task scheduling [1]. Task decomposition divides complex tasks into smaller ones, task allocation assigns tasks to different agents, and task scheduling determines the order in which tasks should be completed. Task allocation is particularly important because agents may have different capabilities, locations, and available resources [1]. In multi-robot systems, some tasks may also require several robots to work together. Bezerra et al. illustrate this problem using emergency response, where multiple firetrucks must cooperate to handle incidents that cannot be managed by one vehicle alone [2].

Task allocation becomes more difficult in dynamic environments, where new tasks may appear at any time and agents have limited information about their surroundings. For example, an emergency incident may require multiple rescue robots, while a smaller task can be completed by a single robot. Traditional task allocation approaches include optimization algorithms, auction-based methods, and predefined rules [1]. More recent approaches use Deep Reinforcement Learning (DRL), which allows agents to learn task allocation strategies through interaction with their environment [3], [2].

Communication is an important part of decentralized task allocation. By sharing information about their observations, selected tasks, and planned actions, agents can make better decisions and coordinate their work. However, communication may become unreliable due to limited network coverage, interference, or message loss. Missing information can lead to repeated task assignments, delays, and problems when several agents need to complete a task together. These challenges highlight the importance of adaptive coordination strategies in dynamic multi-agent environments [4], [5].

The practical part of this paper focuses on how Deep Reinforcement Learning can support cooperation between agents in dynamic task allocation. MARL-MRT^{2}A is used as a case study to examine how task allocation performance develops during training, particularly for tasks requiring different levels of cooperation [2]. The reproduced results are compared across training stages and considered alongside the traditional PCFA approach.

II. Problem Formulation

- Agents and tasks in a dynamic environment.
- Task assignment and coalition formation.
- Challenges of limited information and changing tasks.

III. Deep Learning-Based Task Allocation

- Deep Learning approaches to task allocation.
- Deep Reinforcement Learning and Multi-Agent Reinforcement Learning.
- Learning task selection and coordination strategies.

IV. Use Case and Experimental Setup

- Multi-robot task allocation in emergency response.
- MARL-MRT^{2}A and the GyMRT^{2}A simulation environment.
- Experimental evaluation and comparison with PCFA.

V. Results and Evaluation

- How performance changes during training (comparison of saved model checkpoints).
- Task completion and reward results.
- Comparison with this Deep Learning approach with the classical PCFA.

VI. Discussion

- Advantages and limitations of Deep Learning for task allocation.
- Challenges of applying the approach in real environments.

VII. Conclusion

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